

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3261

3 Credits: 4

Program Core

Source-grounded

Fundamentals of Network

□ To provide basic concepts of Computer Networks from a Network Administrator's

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3261 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3261.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Information Technology
Course code	3261
Course title	Fundamentals of Network
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

39 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide basic concepts of Computer Networks from a Network Administrator's
- point-of-view.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding its Physical Layer. Illustrate basic concepts of Local Area Networks, IP	See official syllabus
CO2	14 Understanding Addressing and Subnetting. Summarize basic concepts of Routing Algorithms and	See official syllabus
CO3	15 Understanding Routing Protocols. Illustrate basic concepts of Transport layer protocols,	See official syllabus
CO4	14 Understanding Application layer protocols and Network Security.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the basic concepts of Computer Networks and its Physical Layer.	7	As specified
Module 2	Subnetting.	13	As specified
Module 3	Summarize basic concept of Routing Algorithms and Routing Protocols	8	As specified
Module 4	protocols and Network Security.	11	As specified

MODULE 1

Outline the basic concepts of Computer Networks and its Physical Layer.

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the basic concepts of Computer Networks and its Physical Layer.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ elements of a computer network and 2 Understanding Network Topologies

elements of a computer network and 2 Understanding Network Topologies is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify elements of a computer network and 2 Understanding Network Topologies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, elements of a computer network and 2 understanding network topologies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What elements of a computer network and 2 Understanding Network Topologies means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്ന elements of a computer network and 2 Understanding Network Topologies എന്നിവയുടെ definition/meaning എഴുതുക. എന്നിവയുടെ components, steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ Illustrate Transmission Mediums.

Illustrate Transmission Mediums. is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Transmission Mediums. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate transmission mediums. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Transmission Mediums. means in the official course context

▼ Outline Types of Networks

Outline Types of Networks is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Types of Networks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline types of networks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Types of Networks means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Outline Types of Networks
definition/meaning .
steps,
application . Technical terms English-
.

▼ Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI

Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize switched networks 2 understanding illustrate the duties of each layer in osi should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

▼ 3 Understanding Model Outline the duties of each layer in

3 Understanding Model Outline the duties of each layer in is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Model Outline the duties of each layer in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding model outline the duties of each layer in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Model Outline the duties of each layer in means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding Model Outline the duties of each layer in CNN definition/meaning CNN . CNN CNN CNN , steps, application CNN CNN CNN . Technical terms English- CNN CNN CNN .

▼ 2 Understanding TCP/IP Model

2 Understanding TCP/IP Model is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding TCP/IP Model using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding tcp/ip model should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding TCP/IP Model means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding TCP/IP Model

പ്രദേശം definition/meaning പ്രദേശം. പ്രദേശം പ്രദേശം പ്രദേശം, steps, application പ്രദേശം പ്രദേശം. Technical terms English- പ്രദേശം പ്രദേശം.

▼ Summarize Network Devices. 2 Understanding Computer Network - Basic Elements - Network Topologies (Bus, Star, Ring and Mesh) - Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) - Types of Networks (LAN, MAN, WAN, PAN and Wireless) - Switched Networks (Circuit Switched, Packet Switched) Protocol Stack - OSI Model - Functions of each layer - TCP/IP Model - Functions of Each Layer - Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and

Summarize Network Devices. 2 Understanding Computer Network - Basic Elements - Network Topologies (Bus, Star, Ring and Mesh) - Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) - Types of Networks (LAN, MAN, WAN, PAN and Wireless) - Switched Networks (Circuit Switched, Packet Switched) Protocol Stack - OSI Model - Functions of each layer - TCP/IP Model - Functions of Each Layer - Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and is an official topic in Module 1 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Network Devices. 2 Understanding Computer Network - Basic Elements - Network Topologies (Bus, Star, Ring and Mesh) -

Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize network devices. 2 understanding computer network – basic elements – network topologies (bus, star, ring and mesh) – transmission medium (twisted pair, co-axial, fiber optic and electromagnetic waves) – types of networks (lan, man, wan, pan and wireless) – switched networks (circuit switched, packet switched) protocol stack – osi model – functions of each layer – tcp/ip model – functions of each layer – difference between osi and tcp/ip models networking devices (nic, hub, modem, bridge, switch, router, gateway and firewall) illustrate basic concepts of local area networks, ip addressing and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models

Aspect	What to prepare
	Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-☐☐ Summarize Network Devices. 2 Understanding Computer Network – Basic Elements – Network Topologies (Bus, Star, Ring and Mesh) – Transmission Medium (Twisted Pair, Co-axial, Fiber Optic and Electromagnetic Waves) – Types of Networks (LAN, MAN, WAN, PAN and Wireless) – Switched Networks (Circuit Switched, Packet Switched) Protocol Stack – OSI Model – Functions of each layer – TCP/IP Model – Functions of Each Layer – Difference between OSI and TCP/IP models Networking Devices (NIC, Hub, Modem, Bridge, Switch, Router, Gateway and Firewall) Illustrate basic concepts of Local Area Networks, IP Addressing and ☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Subnetting.

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Subnetting.
- Official duration: As specified
- Official topic entries captured: 13
- The full source excerpt is preserved below for coverage checking.

▼ 1 Understanding Techniques

1 Understanding Techniques is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Techniques means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding Techniques

പ്രദേശം definition/meaning. പ്രദേശം, steps, application. Technical terms English-ൽ പ്രദേശം.

▼ Summarize Ethernet and its types

Summarize Ethernet and its types is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Ethernet and its types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize ethernet and its types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Ethernet and its types means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നെക്കുറിച്ച് Summarize Ethernet and its types
 എന്താണ് definition/meaning എന്താണ്. എന്താണ്
 എന്താണ്, steps, application എന്താണ് എന്താണ്. Technical terms
 English-ൽ എന്താണ് എന്താണ്.

▼ Outline ARP and RARP

Outline ARP and RARP is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline ARP and RARP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline arp and rarp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline ARP and RARP means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ARP Outline ARP and RARP ഏകദേശം
 ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps,
 application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-ഏകദേശം
 ഏകദേശം.

▼ Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP

Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize ip address and ip 1 understanding addressing outline classful ip addressing and ip should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP definition/meaning. , steps, application . Technical terms English- .

▼ 1 Understanding Address Classes.

1 Understanding Address Classes. is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Address Classes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding address classes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Address Classes. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding Address Classes.

പ്രധാനപ്പെട്ട പദങ്ങൾക്ക് definition/meaning നൽകുക. പദങ്ങൾക്ക് പദപദങ്ങൾ, steps, application നൽകുക. Technical terms English-ൽ പദപദങ്ങൾ നൽകുക.

▼ Summarize IPv4 Addressing Types

Summarize IPv4 Addressing Types is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize IPv4 Addressing Types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize ipv4 addressing types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize IPv4 Addressing Types means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
 Summarize IPv4 Addressing Types
 definition/meaning. , steps, application . Technical terms
 English- .

▼ Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4

Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize ipv4 reserved addresses 1 understanding illustrate packet flow in an ipv4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Summarize IPv4 Reserved Addresses 1 Understanding Illustrate Packet Flow in an IPv4 definition/meaning . . . , steps, application Technical terms English-

▼ 1 Understanding Network

1 Understanding Network is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Network using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding network should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Network means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പാഠ്യ 1 Understanding Network പദപരിചയം
 പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps,
 application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം
 പദപരിചയം.

▼ Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful

Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize ipv4 fragmentation 1 understanding outline limitations of classful should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Summarize IPv4 Fragmentation 1 Understanding Outline Limitations of Classful definition/meaning . steps, application . Technical terms English- .

▼ 1 Understanding Addressing

1 Understanding Addressing is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Addressing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding addressing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Addressing means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding Addressing

പ്രദേശീയമായി നിർവ്വചിക്കുക definition/meaning നിർവ്വചിക്കുക. നിർവ്വചിക്കുക നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless

Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate subnet mask and subnetting 2 understanding summarize supernetting and classless should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Illustrate Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നു കാണിക്കുക Subnet Mask and Subnetting 2 Understanding Summarize Supernetting and Classless എന്നിവയുടെ അർത്ഥം/definition/meaning എഴുതുക. എന്നിവയുടെ ഘട്ടങ്ങൾ, steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ 1 Understanding Inter-Domain Routing

1 Understanding Inter-Domain Routing is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Inter-Domain Routing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding inter-domain routing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Inter-Domain Routing means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-1 Understanding Inter-Domain Routing
 പരസ്പരം വിവിധ ഡോമൈനുകളിലെ നെറ്റ്‌വർക്കുകൾ തമ്മിൽ കമ്മ്യൂണിക്കേഷൻ സാധ്യമാക്കുന്നതിനായി ഉപയോഗിക്കുന്ന പ്രക്രിയയാണ് ഇന്റർ-ഡോമൈൻ റൂട്ടിംഗ്. ഇതിൽ ഉൾപ്പെടുന്നത്, steps, application എന്നിവയാണ്. Technical terms English-ൽ പരാമർശിക്കുന്നു.

▼ Outline Variable Length Subnet Mask 1 Understanding Local Area Network - Types of Network Links - Medium Access Control Techniques - Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) - Ethernet - Types of Ethernet ARP and RARP - Address Resolution Protocol - Reverse Address Resolution Protocol IP Addressing - IP Address - IPv4 - IPv6 - Classful Addressing - IPv4 Address Classes (Class A, B, C, D and E) - IPv4 Addressing Types (Unicast, Broadcast and Multicast) - IPv4 Reserved Address (Loopback, Zero and Private Addresses) - Packet Flow in IPv4 Network - IPv4 Fragmentation - Limitations of Classful Addressing - Subnet Mask - Subnetting - Supernetting - Classless Inter-Domain Routing - Variable Length Subnet Mask

Outline Variable Length Subnet Mask 1 Understanding Local Area Network - Types of Network Links - Medium Access Control Techniques - Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) - Ethernet - Types of Ethernet ARP and RARP - Address Resolution Protocol - Reverse Address Resolution Protocol IP Addressing - IP Address - IPv4 - IPv6 - Classful Addressing - IPv4 Address Classes (Class A, B, C, D and E) - IPv4 Addressing Types (Unicast, Broadcast and Multicast) - IPv4 Reserved Address (Loopback, Zero and Private Addresses) - Packet Flow in IPv4 Network - IPv4 Fragmentation - Limitations of Classful Addressing - Subnet Mask - Subnetting - Supernetting - Classless Inter-Domain Routing - Variable Length Subnet Mask is an official topic in Module 2 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline variable length subnet mask 1 understanding local area network – types of network links – medium access control techniques – random access techniques (aloha, csma, csma/cd, csma/ca) – ethernet – types of ethernet arp and rarp – address resolution protocol – reverse address resolution protocol ip addressing – ip address – ipv4 – ipv6 – classful addressing – ipv4 address classes (class a, b, c, d and e) – ipv4 addressing types (unicast, broadcast and multicast) – ipv4 reserved address (loopback, zero and private addresses) – packet flow in ipv4 network – ipv4 fragmentation – limitations of classful addressing – subnet mask – subnetting – supernetting – classless inter-domain routing – variable length subnet mask should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline Variable Length Subnet Mask 1 Understanding Local Area Network – Types of Network Links – Medium Access Control Techniques – Random Access Techniques (ALOHA, CSMA, CSMA/CD, CSMA/CA) – Ethernet – Types of Ethernet ARP and RARP – Address Resolution Protocol – Reverse Address Resolution Protocol IP Addressing – IP Address – IPv4 – IPv6 – Classful Addressing – IPv4 Address Classes (Class A, B, C, D and E) – IPv4 Addressing Types (Unicast, Broadcast and Multicast) – IPv4 Reserved Address (Loopback, Zero and Private Addresses) – Packet Flow in IPv4 Network – IPv4 Fragmentation – Limitations of Classful Addressing – Subnet Mask – Subnetting – Supernetting – Classless Inter-Domain Routing – Variable Length Subnet Mask □□□□□□□□□□ □□□□□ definition/meaning

പ്രവർത്തനങ്ങൾ. പദങ്ങൾ പദങ്ങൾ, steps, application പദങ്ങൾ
പ്രവർത്തനങ്ങൾ പദങ്ങൾ. Technical terms English-പദങ്ങൾ പദങ്ങൾ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize basic concept of Routing Algorithms and Routing Protocols

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize basic concept of Routing Algorithms and Routing Protocols
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding Algorithms Summarize Classification of Routing

2 Understanding Algorithms Summarize Classification of Routing is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Algorithms Summarize Classification of Routing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding algorithms summarize classification of routing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 2 Understanding Algorithms Summarize Classification of Routing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding Algorithms Summarize Classification of Routing ഏതെങ്കിലും രീതിയിൽ definition/meaning ഉൾപ്പെടുത്തുക. ഏതെങ്കിലും രീതിയിൽ, steps, application ഉൾപ്പെടുത്തുക. Technical terms English-ൽ ഉൾപ്പെടുത്തുക.

▼ 2 Understanding Algorithms

2 Understanding Algorithms is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Algorithms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding algorithms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Algorithms means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding Algorithms

പ്രശ്നത്തിന്റെ പദപദാർത്ഥം definition/meaning നൽകുക. പ്രശ്നത്തിന്റെ പദപദാർത്ഥം, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

▼ Outline Routing Algorithm Metrics

Outline Routing Algorithm Metrics is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Routing Algorithm Metrics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline routing algorithm metrics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Routing Algorithm Metrics means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഔ Outline Routing Algorithm Metrics
 ഔഔഔഔഔഔഔഔ ഔഔഔ definition/meaning ഔഔഔഔഔഔഔഔ. ഔഔഔഔ ഔഔഔഔ
 ഔഔഔഔഔ, steps, application ഔഔഔഔ ഔഔഔഔഔഔഔ ഔഔഔഔ. Technical terms
 English-ഔ ഔഔഔഔ ഔഔഔഔഔഔഔഔ.

▼ Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its

Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize internet architecture 1 understanding outline routing protocols and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its definition/meaning. definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 2 Understanding classifications. Summarize Routing Information

2 Understanding classifications. Summarize Routing Information is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding classifications. Summarize Routing Information using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding classifications. summarize routing information should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 2 Understanding classifications. Summarize Routing Information means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding classifications. Summarize Routing Information definition/meaning. 2 Understanding classifications, steps, application. Technical terms English- Malayalam.

▼ 2 Understanding Protocol Illustrate Open Shortest Path First

2 Understanding Protocol Illustrate Open Shortest Path First is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Protocol Illustrate Open Shortest Path First using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding protocol illustrate open shortest path first should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Protocol Illustrate Open Shortest Path First means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding Protocol Illustrate Open Shortest Path First definition/meaning. steps, application. Technical terms English-

▼ 2 Understanding Routing Protocol

2 Understanding Routing Protocol is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Routing Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding routing protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Routing Protocol means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding Routing Protocol
 പരീക്ഷയ്ക്കായി definition/meaning നൽകുക. പരീക്ഷയിൽ പരീക്ഷിക്കുക
 പരീക്ഷിക്കുക, steps, application നൽകുക പരീക്ഷിക്കുക പരീക്ഷിക്കുക. Technical terms
 English-ൽ പരീക്ഷിക്കുക പരീക്ഷിക്കുക.

▼ Summarize Border Gateway Protocol 2 Understanding IP Routing - Routing Algorithm - Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) - Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) - Internet Architecture Routing Protocols - Classification - Routing Information Protocol - Open Shortest Path First - Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer

Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer is an official topic in Module 3 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path

Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize border gateway protocol 2 understanding ip routing – routing algorithm – classification (static vs dynamic, single-path vs multi- path, intra-domain vs inter-domain, link-state vs distance vector, host intelligent vs router intelligent) – routing algorithm metrics (path length, delay, bandwidth, load, communication cost and reliability) – internet architecture routing protocols – classification – routing information protocol – open shortest path first – border gateway protocol. illustrate basic concepts of transport layer protocols, application layer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Border Gateway Protocol 2 Understanding IP Routing – Routing Algorithm – Classification (Static vs Dynamic, Single-Path vs Multi- Path, Intra-Domain vs Inter-Domain, Link-State vs Distance Vector, Host Intelligent vs Router Intelligent) – Routing Algorithm Metrics (Path Length, Delay, Bandwidth, Load, Communication Cost and Reliability) – Internet Architecture Routing Protocols – Classification – Routing Information Protocol – Open Shortest Path First – Border Gateway Protocol. Illustrate basic concepts of Transport layer protocols, Application layer means in the official course context
Study lens	Principle → components or stages → result → application

MODULE 4

protocols and Network Security.

This module is presented from the official syllabus scope for Fundamentals of Network. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: protocols and Network Security.
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

▼ Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol

Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate ports and sockets 1 understanding outline transmission control protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇല്ലാത്ത Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol ഇല്ലാത്ത definition/meaning ഇല്ലാത്ത. ഇല്ലാത്ത ഇല്ലാത്ത ഇല്ലാത്ത, steps, application ഇല്ലാത്ത ഇല്ലാത്ത ഇല്ലാത്ത. Technical terms English-ഇല്ലാത്ത ഇല്ലാത്ത ഇല്ലാത്ത.

▼ 2 Understanding and User Datagram Protocols

2 Understanding and User Datagram Protocols is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding and User Datagram Protocols using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding and user datagram protocols should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding and User Datagram Protocols means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding and User Datagram Protocols
 definition/meaning .
 , steps, application . Technical terms English-
 .

▼ Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control

Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize hypertext transfer protocol 2 understanding outline simple mail transfer control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

▼ 1 Understanding Protocol

1 Understanding Protocol is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Protocol means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-1 Understanding Protocol
 definition/meaning . steps, application . Technical terms English-
 .

▼ Illustrate File Transfer Protocol

Illustrate File Transfer Protocol is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate File Transfer Protocol using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate file transfer protocol should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate File Transfer Protocol means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓഫ് Illustrate File Transfer Protocol

ഓഫ് definition/meaning ഓഫ്. ഓഫ് ഓഫ് ഓഫ്, steps, application ഓഫ് ഓഫ്. Technical terms English-ഓഫ് ഓഫ്.

▼ Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration

Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize domain name server 1 understanding outline dynamic host configuration should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Summarize Domain Name Server 1 Understanding Outline Dynamic Host Configuration definition/meaning . steps, application Technical terms English- Malayalam.

▼ 1 Understanding Protocol Summarize Network Security Needs,

1 Understanding Protocol Summarize Network Security Needs, is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Protocol Summarize Network Security Needs, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding protocol summarize network security needs, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Protocol Summarize Network Security Needs, means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 1 Understanding Protocol Summarize Network Security Needs, definition/meaning, steps, application, Technical terms English- Malayalam.

▼ 2 Understanding History and Four Pillars Illustrate Network Security Terms and

2 Understanding History and Four Pillars Illustrate Network Security Terms and is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding History and Four Pillars Illustrate Network Security Terms and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding history and four pillars illustrate network security terms and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 2 Understanding History and Four Pillars Illustrate Network Security Terms and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding History and Four Pillars Illustrate Network Security Terms and definition/meaning. steps, application. Technical terms English- .

▼ 1 Understanding Security Components Summarize Types of Attacks and

1 Understanding Security Components Summarize Types of Attacks and is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Security Components Summarize Types of Attacks and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding security components summarize types of attacks and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 1 Understanding Security Components Summarize Types of Attacks and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 1 Understanding Security Components Summarize Types of Attacks and definition/meaning. principle, components or stages, steps, application appropriate. Technical terms English- Malayalam.

▼ 1 Understanding Known Security Attacks

1 Understanding Known Security Attacks is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Known Security Attacks using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding known security attacks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Known Security Attacks means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-1 Understanding Known Security Attacks
 പരിചയപ്പെടുത്തൽ definition/meaning പരിചയപ്പെടുത്തൽ. പരിചയപ്പെടുത്തൽ
 പരിചയപ്പെടുത്തൽ, steps, application പരിചയപ്പെടുത്തൽ പരിചയപ്പെടുത്തൽ. Technical terms
 English-ൽ പരിചയപ്പെടുത്തൽ പരിചയപ്പെടുത്തൽ.

▼ **Outline** **Firewalls** **1 Understanding TCP and UDP - Ports - Sockets - User Datagram Protocols - Transmission Control Protocol - Compare TCP and UDP Application Layer Protocols - Hypertext Transfer Protocol - Simple Mail Transfer Control Protocol - File Transfer Protocol - Domain Name Server - Dynamic Host Configuration Protocol Network Security - Needs - History - Four Pillars - Security Terms - Security Components - Types of Attacks - Known Security Attacks - Firewalls** **Text / Reference:** **T/R Book Title / Author** **Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016** **Online Resources: Sl.No. Website Link**
1 https://www.tutorialspoint.com/data_communication_computer_network/index1.htm **2** <https://www.javatpoint.com/computer-network-tutorial> **3** <https://www.geeksforgeeks.org/computer-network-tutorials/>

Outline **Firewalls** **1 Understanding TCP and UDP - Ports - Sockets - User Datagram Protocols - Transmission Control Protocol - Compare TCP and UDP Application Layer Protocols - Hypertext Transfer Protocol - Simple Mail Transfer Control Protocol - File Transfer Protocol - Domain Name Server - Dynamic Host Configuration Protocol Network Security - Needs - History - Four Pillars - Security Terms - Security Components - Types of Attacks - Known Security Attacks - Firewalls** **Text / Reference:** **T/R Book Title / Author** **Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016** **Online Resources: Sl.No. Website Link**
https://www.tutorialspoint.com/data_communication_computer_network/index1.htm **2** <https://www.javatpoint.com/computer-network-tutorial> **3** <https://www.geeksforgeeks.org/computer-network-tutorials/> is an official topic in Module 4 of Fundamentals of Network. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link
https://www.tutorialspoint.com/data_communication_computer_network/index.
 1 tm 2 <https://www.javatpoint.com/computer-network-tutorial> 3
<https://www.geeksforgeeks.org/computer-network-tutorials/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline firewalls 1 understanding tcp and udp – ports – sockets – user datagram protocols – transmission control protocol – compare tcp and udp application layer protocols – hypertext transfer protocol – simple mail transfer control protocol – file transfer protocol – domain name server – dynamic

host configuration protocol network security – needs – history – four pillars – security terms – security components – types of attacks – known security attacks – firewalls text / reference: t/r book title / author narasimha karumanchi, dr. a. damodaram and dr. m. sreenivasa rao. elements of t1 computer networking: an integrated approach. careermonk publications, 2014. todd lammle, ccna routing and switching complete study guide, 2nd edition, r1 sybex (wiley) publication, 2016 online resources: sl.no. website link

[https://www.tutorialspoint.com/data_communication_computer_network/index.h](https://www.tutorialspoint.com/data_communication_computer_network/index.htm)
1 tm 2 <https://www.javatpoint.com/computer-network-tutorial> 3

<https://www.geeksforgeeks.org/computer-network-tutorials/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link https://www.tutorialspoint.com/data_communication_computer_network/index 1 tm 2 https://www.javatpoint.com/computer-network-tutorial 3 https://www.geeksforgeeks.org/computer-network-tutorials/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally

supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Outline Firewalls 1 Understanding TCP and UDP – Ports – Sockets – User Datagram Protocols – Transmission Control Protocol – Compare TCP and UDP Application Layer Protocols – Hypertext Transfer Protocol – Simple Mail Transfer Control Protocol – File Transfer Protocol – Domain Name Server – Dynamic Host Configuration Protocol Network Security – Needs – History – Four Pillars – Security Terms – Security Components – Types of Attacks – Known Security Attacks – Firewalls Text / Reference: T/R Book Title / Author Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. Elements of T1 Computer Networking: An Integrated Approach. CareerMonk Publications, 2014. Todd Lammle, CCNA Routing and Switching Complete Study Guide, 2nd Edition, R1 Sybex (Wiley) Publication, 2016 Online Resources: Sl.No. Website Link
1 tm 2 <https://www.javatpoint.com/computer-network-tutorial> 3 <https://www.geeksforgeeks.org/computer-network-tutorials/> definition/meaning . steps, application . Technical terms English- .

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain elements of a computer network and 2 Understanding Network Topologies. (3 marks)

► Define or explain Illustrate Transmission Mediums.. (3 marks)

► Define or explain Outline Types of Networks. (3 marks)

► Define or explain Summarize Switched Networks 2 Understanding Illustrate the duties of each layer in OSI. (3 marks)

Module 2

► Define or explain 1 Understanding Techniques. (3 marks)

► Define or explain Summarize Ethernet and its types. (3 marks)

► Define or explain Outline ARP and RARP. (3 marks)

► Define or explain Summarize IP Address and IP 1 Understanding Addressing Outline Classful IP Addressing and IP. (3 marks)

Module 3

► **Define or explain 2 Understanding Algorithms Summarize Classification of Routing. (3 marks)**

► **Define or explain 2 Understanding Algorithms. (3 marks)**

► **Define or explain Outline Routing Algorithm Metrics. (3 marks)**

► **Define or explain Summarize Internet Architecture 1 Understanding Outline Routing Protocols and its. (3 marks)**

Module 4

► **Define or explain Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol. (3 marks)**

► **Define or explain 2 Understanding and User Datagram Protocols. (3 marks)**

► **Define or explain Summarize Hypertext Transfer Protocol 2 Understanding Outline Simple Mail Transfer Control. (3 marks)**

► **Define or explain 1 Understanding Protocol. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **elements of a computer network and 2 Understanding Network Topologies**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **1 Understanding Techniques**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding Algorithms Summarize Classification of Routing**.

3-mark explanation: Explain one principle, process or comparison from

Module 4

1-mark definition: State the meaning of **Illustrate Ports and Sockets 1 Understanding Outline Transmission Control Protocol**.

3-mark explanation: Explain one principle, process or comparison from

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No. Website Link
- https://www.tutorialspoint.com/data_communication_computer_network/index.h
- <https://www.javatpoint.com/computer-network-tutorial>
- <https://www.geeksforgeeks.org/computer-network-tutorials/>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 3261. Verify institutional updates against the current official curriculum.